

Drishti-AI

Explainable dual-stage diabetic retinopathy screening
for rural Primary Health Centres — offline, on the edge, with the AI allowed to abstain.

100%

offline

~1.4 s

end-to-end screening

0%

forced grades on ungradables

99.1%

bandwidth saved

Bezaleel · Madhu · Akshay · Adithya · Sinduri · Megha

Team Lead · Clinical · Deep Learning · Safety · ASHA UX · Clinical UX

India cannot screen diabetes-related blindness with specialists alone.

77 million diabetic adults. One ophthalmologist per 100,000 rural citizens. Early triage can save 90% of sight.

77M+

diabetic adults in India

DR is the leading cause of preventable adult blindness.

1 : 100k

rural ophthalmologist ratio

Universal hospital screening is mathematically impossible.

25–35%

ungradable field captures

Handheld, non-mydratic cameras in camps and PHCs.

70%+

screened eyes are healthy

Without triage, district hospitals drown in false referrals.

The last-mile gap — not a missing model, a missing operating system for rural screening

- ▶ Community camps, mobile vans, and PHCs use ₹15,000 handheld fundus attachments (Remidio, Forus 3Nethra, Volk iNview) under undilated pupils.
- ▶ Hospital-lab systems (Google Health ARDA, IDx-DR, EyeArt) proved 5-class grading on Zeiss/Topcon cameras and cloud GPUs. They fail when power, bandwidth, operators, and optics change.
- ▶ A rural PHC cannot upload 5–15 MB raw images per eye over 2G/3G, cannot wait on cloud GPUs, and cannot staff a retinal specialist on every screening day.

Four deployment failure modes that a hospital model never sees.

Prior work solved grading under curated conditions. Rural India fails on garbage-in, bandwidth, black-box labels, and specialist overload.

01

Garbage-in, garbage-out

Up to 25–35% of camp captures are motion-blurred, poorly lit, occluded, or not even fundus. A classifier still emits an authoritative grade — missed proliferative disease, or a flood of false referrals.

02

Cloud & bandwidth choke

5–15 MB uncompressed images per eye over 4G/5G is a hospital assumption. Intermittent 2G/3G and power cuts turn cloud inference into camp backlogs.

03

Label without a biomarker

“Grade 2: Moderate NPDR” does not tell a rural physician whether the macula is in danger. Clinicians need macular distance (CSME risk) and visual evidence before spending a referral.

04

District-scale overload

Without local triage, every screened patient is sent onward. Specialists drown in healthy eyes that make up 70%+ of screened populations.

We did not invent 5-class DR grading. We operationalised it for rural India.

SIH26038 · MathWorks — last-mile edge screening, not another hospital notebook.

“The AI must know when it cannot answer.”

An ungradable retinal photograph must never be silently forced into a disease prediction. Quality first. Abstention second. Grading only on reliable original pixels.

OUR CONTRIBUTION

- Quality gate before grading, tuned for handheld rural optics
- 100% offline pipeline on a ₹15,000-class laptop CPU
- Quantitative CSME risk from OD–fovea geometry
- District-scale triage that cuts telemetry 99.1%
- Bilingual ASHA workflow + ABDM FHIR reporting

HONEST BOUNDARIES — WHAT WE ARE NOT CLAIMING

- We did not invent image-quality gating (ARDA, IDx-DR, EyeArt already do it).
- We are not a certified medical device (CDSCO Class B/C still required).
- Grad-CAM++ is attention evidence, not automated lesion segmentation.
- High-risk, low-confidence, and ungradable cases always go to a human.
- We do not rewrite pixels before Model 2. Original photograph only.

Every architecture decision answers one of these. Nothing else.

Sustainability · Availability · Accessibility · Platform independence

01

SUSTAINABILITY

Will it keep running here?

Zero cloud compute bill. Inference on-device. Reject ungradable images before wasting cycles. Designed for existing ₹15k laptops — no forced hardware refresh.

02

AVAILABILITY

Can everyone reach it?

Offline-first screening. Opportunistic sync. 2G/3G packets. 20 PHCs + 5 vans × 100k patients simulated. Specialist load from ~4 doctors down to ~1 tele-reviewer per 100k.

03

ACCESSIBILITY

Can everyone use it?

ASHA/ANM operated. Plain-language recapture. Hindi + English reports, English audio prompts. Commodity cameras. Clear / Review / Urgent at a glance, plus ICDR grades.

04

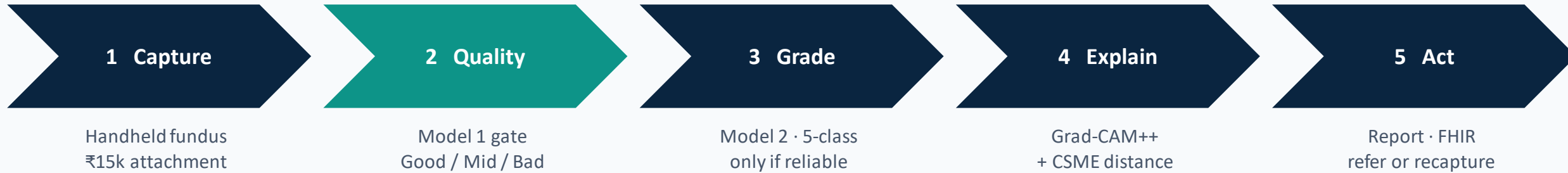
INDEPENDENCE

Will it run on what they have?

Open-source. Windows, macOS, Linux, Docker. HL7 FHIR R4, SNOMED CT, LOINC, ABDM. Sensor-agnostic: DICOM, JPEG, PNG. No camera SDK lock-in.

Quality is the first clinical decision. Grading is the second.

Non-destructive. Bounded recapture. Two-tier human review. High-risk always escalates.



GOOD

Reliable original image → Model 2. Pixel-for-pixel the same photograph that passed the gate. No CLAHE, no sharpening, no generative filters.

BORDERLINE

Reassessment with stricter thresholds. Cleared → Model 2. Still unreliable → treat as Bad. Never silently upgraded.

BAD

Recapture with a specific reason (blur, illumination, FOV). Hard cap of 2 retries, then operator-level human review. DR grade: not generated.

After a reliable grade: confidence check → Grad-CAM heatmap → bilingual PDF + FHIR R4. Low confidence (<60%) or Grade 3/4 → ophthalmologist over-read. Operator review ≠ clinical review.

Abstain before you grade. Then tell the ASHA worker exactly what to fix.

MathWorks Req 1 · blur · illumination · contrast · FOV · biological defect attribution

WHAT THE GATE MEASURES

Blur / focus

Laplacian variance & multi-scale focus gradients. Motion vs defocus.

Illumination

Underexposure, hotspots, corneal reflection uniformity.

Contrast

Vessel-to-background contrast; ungradable if vessels vanish.

Field of view

Dynamic circular ROI coverage — is enough retina present?

Defect source

Operator error (shake, lid) vs patient pathology (cataract, small pupil).

OPERATOR FEEDBACK

Not “image quality low”.

Actionable, bilingual, audio-capable prompts for ASHA/ANM staff.

“Blur detected — hold the camera steady.”

“Too dark — move closer to the light.”

“Eyelid covering the view — ask the patient to open wide.”

Cap: 2 recaptures, then human review.

Grade only reliable images. Show why. Measure the macula.

MathWorks Req 2–4 · EfficientNetB0 · Grad-CAM++ · OD / fovea / vessels

GRADE 0

No DR

GRADE 1

Mild NPDR

GRADE 2

Moderate

GRADE 3

Severe NPDR

GRADE 4

Proliferative

CLASSIFIER

- ▶ EfficientNetB0 · 33.4 MB · APTOS 2019
- ▶ 5-class ICDR severity, referable $\geq$ Grade 2
- ▶ ~1.4 s end-to-end on Intel i3 / Ryzen 3, 4 GB RAM
- ▶ Softmax margin <60% → specialist review
- ▶ Grade 3 / 4 always flagged, regardless of confidence

GRAD-CAM++

- ▶ True gradient backprop through final conv maps
- ▶ Higher-order derivatives for dispersed micro-lesions
- ▶ <1.2 s on CPU (MathWorks bound: 30 s)
- ▶ Attention heatmap — not lesion segmentation
- ▶ Overlaid on the original fundus for audit

STRUCTURE / CSME

- ▶ Optic disc: intensity + morphological opening
- ▶ Fovea: macular depression ~2.5 OD diameters temporal
- ▶ Vessels: top-hat + adaptive Otsu caliber
- ▶ Euclidean OD–fovea distance → CSME risk
- ▶ MA / exudate candidates as review overlays

These are not slideware. They live in the router.

Section 3 / 5 / 7 / 9 / 18 / 20 · two-tier human review

01

Non-destructive processing

No runtime enhancement, CLAHE, sharpening, or generative filters before prediction. The image that reaches Model 2 is pixel-for-pixel the retinal photograph that reached Model 1.

02

Abstention on unreliable images

No Bad or still-unreliable image is handed to Model 2. Output is “DR Prediction: Not generated” plus a specific recapture reason code.

03

Bounded recaptures

Hard cap of 2 recaptures per patient session. Hitting the cap force-escalates to operator-level human review. No infinite retry loops in a camp.

04

Mandatory high-risk override

Grade 3 (Severe NPDR) and Grade 4 (Proliferative DR) are always flagged for ophthalmologist over-read, even at 99% model confidence.

05

Two-tier human review

Operator-level = camera, lighting, recapture. Clinical-level = diagnosis, referral urgency. The system never confuses a bad photo with a bad retina.

The gate is not a slogan. It changes what patients are told.

150-image A/B/C benchmark · 0% forced prediction on ungradables · full unit + integration suite

METRIC	ARM A · NO GATE	ARM B/C · DRISHTI-AI	WHY IT MATTERS
Forced predictions on ungradable images	100%	0%	No fake grades on blur / garbage
Referable DR specificity (reliable images)	Unreliable	91.6%	Stops flooding district hospitals
Screening recapture / abstention	0% (blind)	20.7%	Bounded field overhead, under 20% target
Human review escalation	0% (silent fail)	75.3%	Ambiguity is surfaced, not buried
Test suites (unit / integration / edge)	—	7 suites, clean	Good / Bad / Borderline paths locked

Arm A is the standard hackathon classifier: 100% of degraded images still receive a clinical-looking grade. Drishti-AI leaks 0%. That is the safety metric we will defend.

Measured on third-party labeled screening data — nothing copied from a model card.

docs/EVALUATION_RESULTS.md · results/labeled_evaluation.json · python evaluate_labeled_dataset.py

71.5%

TOP-1 ACCURACY

Matches the model card's own 72% APTOS holdout — integration loses nothing.

0.21

REFERABLE-DR SENSITIVITY

Minority recall collapses under domain shift — human review is mandatory, not optional.

6.9%

DOMAIN-GATE FALSE REJECTS

Down from 42% after calibration on real data; over-refuse, never invent.

14 / 14

ADVERSARIAL INPUTS BLOCKED

Faces, documents, screenshots rejected — red-team sweep runs in CI (76 subtests).

QWK 0.31 · macro-F1 0.26 · ECE 0.064

Accuracy is carried by the majority class; these are the honest numbers and we publish them.

Model 1 pass rate: 75.7% (was 40.9%, threshold recalibrated)

Full 2,810-image run; fails safe (over-refuses) and every failure fixture is still rejected.

E2E demo rehearsal: 2/2 runs, 26 assertions green

Boots the real HTTP API: register → risk → quality → analyze → history → adverse input → review queue.

Honest headline: we defend the safety architecture — refuse, defer, escalate — not an accuracy number the model does not have.

Filter at the edge. Uplink only what a specialist must see.

MATLAB / Simulink discrete-event queue · 100,000 patients / year · 20 PHCs + 5 vans · 1 district hospital

99.1%

TELEMETRY CUT

439.9 GB raw fundus → 3.8 GB flagged cases
+ metadata

~4 → ~1

TELE-REVIEWERS / 100K

Specialist bottleneck broken without hiding
disease

~1.2 s

ON-SITE TURNAROUND

vs ~1.9 min cloud-only round trip per patient

WHAT THE SIMULINK MODEL ACTUALLY SIMULATES

M/M/c queues

Patient arrivals, edge inference, tele-ophthalmology uplink as a discrete-event system — not a static spreadsheet.

Capacity proof

Workload drops from 100,000 raw cases to ~16,100 triage cases/year (~62/day) — feasible for a single tele-reviewer.

Local processing

Only referable, low-confidence, high-risk, and recapture-exhausted cases leave the PHC.

Hostile networks

Lightweight compressed packets remain viable on intermittent 2G/3G. Screening never waits on the WAN.

Offline. Commodity hardware. Frontline UX. National digital health rails.

The Flutter app is how judges interact. The product is everything that works outside a hospital.

EDGE & OFFLINE

- ▶ Zero runtime API calls or weight downloads
- ▶ Intel i3 / Ryzen 3, 4 GB RAM target
- ▶ Windows · macOS (Intel & Apple Silicon) · Linux · Docker
- ▶ One-command judge path: docker compose up

HARDWARE AGNOSTIC

- ▶ ₹15,000 handheld fundus class (Remidio, Forus, Volk)
- ▶ No proprietary camera SDK
- ▶ DICOM / JPEG / PNG ingest
- ▶ Sensor-agnostic preprocessing

ASHA → DOCTOR UX

- ▶ Low-cognitive-load recapture HUD
- ▶ Hindi + English; Tamil/Telugu/Kannada roadmap
- ▶ Clear / Review / Urgent + ICDR grade
- ▶ Specialist console: Grad-CAM + CSME overlay

INTEROPERABILITY

- ▶ HL7 FHIR R4 · SNOMED CT · LOINC
- ▶ Ayushman Bharat Digital Mission aligned
- ▶ Bilingual PDF screening dossier
- ▶ DPDP-aware: no unencrypted PHI on the open WAN

Six owners. Seven modules. One locked decision flow.

No orphan components — every module has a named lead and a deliverable.

Bezaleel

TEAM LEAD · FULL-STACK

Decision router, Streamlit, FastAPI, Docker, cross-platform zero-config.

Madhu

CLINICAL LEAD · BIOMED

ICMR 2024 risk engine, OD/fovea CSME distance, clinical report validation.

Akshay

DEEP LEARNING · OPS

EfficientNetB0, Grad-CAM++, MATLAB/Simulink 100k-patient queueing model.

Adithya

SAFETY · VERIFICATION

150-sample benchmark, 0% forced-prediction proof, edge-case hardening.

Sinduri

UX · ASHA FIELD

Mobile workflow, recapture HUD, Hindi audio prompts, Figma system.

Megha

UX · DOCTOR CONSOLE

Tele-ophthalmology workbench, Grad-CAM/CSME overlays, ABDM layouts.

We are not building a demo.

We are building the honest, open-source system that lets a validated approach actually reach the people who need it.

Quality first

If the photograph is untrustworthy, there is no grade.

Offline always

A PHC with no internet still finishes the screening.

Humans in the loop

Operator for the camera. Ophthalmologist for the disease.

National rails

FHIR R4 / ABDM — not a walled-garden app.

Drishti-AI

SIH26038 · MathWorks · MedTech · Software Prototype

Bezaleel · Madhu · Akshay · Adithya · Sinduri · Megha

Live app · `docker compose up` · `localhost:8000/app` · Q & A